

Statement of Work (SOW)
Replace Air Storage System, X-31I

1. General

a. The contractor shall replace the existing air storage system (hereinafter referred to as “the system”) with new one. The system is installed at X-31I Inside Machine Shop, Bldg. A-48, Ship Repair Facility and Japan Regional Maintenance Center (SRF-JRMC), US Naval Base Yokosuka.

b. The system shall be used for storage of 6,000 psig and 3,000 psig breathing air.

c. Unless otherwise specified herein, the contractor shall furnish all labor, materials, equipment, transportation, and supervision necessary to accomplish the work.

d. Order of Precedence.

In the event of a conflict between the text of this document and the references or enclosures cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

e. General Safety

All machine parts, components, mechanisms, and assemblies furnished on the unit shall comply with all specific requirements of “OSHA Safety and Health Standard (29CFR1910), General Industry. (Reference (a), (b)). After January 2, 1990, whenever replacement or major repair, renovation or modification of a machine or equipment is performed, and whenever new machines or equipment are installed, energy isolating devices for such machine or equipment shall be designed to accept a lockout device. (Reference (b))

2. Reference

The contractor shall refer to the following standard documents to accomplish the work to the extent specified herein.

(a) 29 CFR 1910.212 – Occupational Safety and Health Standards Subpart O Machinery and Machine Guarding

(b) 29 CFR 1910.147 - Occupational Safety and Health Standards Subpart J General Environmental Controls.

(c) CGA G-7.1 - Commodity Specification for Air

(d) SS521-AG-PRO-010 - U.S. Navy Diving Manual

(e) ASTM G93 - Standard Guide for Cleanliness Levels and Cleaning Methods for Materials and Equipment Used in Oxygen-Enriched Environments

(f) MIL-STD-1622 - Standard Practice for Cleaning of Shipboard Compressed Air Systems

3. Enclosure

The contractor shall refer to the following enclosures to accomplish the work to the extent specified herein.

(1) Connection Diagram (Current)

(2) Connection Diagram (After Replacement)

4. Minimum requirements

a. Unless otherwise specified, stated values represent the minimum.

b. The air provided from existing two (2) High Pressure Air Compressors (HPAC) is maintained as follows to meet Diver's Breathing Air in U.S. Navy Diving Manual, Grade L SCBA Air in CGA G-7.1, and Grade E SCUBA Air in CGA G-7.1.

Constituent	Specification
Percent Oxygen, Balance Predominately Nitrogen	20-22 %
Carbon Dioxide	1,000 ppm (max)
Carbon Monoxide	10 ppm (max)
Odor and taste	Not objectionable
Water by dew point (degrees F at 1 ATM ABS) or By moisture content (ppm or mg/L)	-65°F or 24 ppm or 0.019 mg/L (max)
Total Volatile Organic Compounds (in methane equivalents)	25 ppm (max)
Condensed Oil and other Particulates	0.005 mg/L (max)

c. Unless otherwise specified, gas-wetted surfaces of components including, but not limited to, seamless stainless steel tubes, tube fittings, valves, and gauges, shall be cleaned to meet the requirements of Level C of ASTM G93 or better, and then covered by Foreign Material Exclusion (FME) closures such as covers, caps, plugs until immediately prior to assembly.

d. Pressure

(1) Working pressure before the replacement is 5,000 psig and 3,000 psig in accordance with Enclosure (1).

(2) Working pressure after the replacement shall be 6,000 psig and 3,000 psig in accordance with Enclosure (2).

e. All components shall be installed in accordance with the manufacturer's technical instruction.

f. Cylinders

(1) Cylinders shall comply with ASME standard.

(2) Cylinders shall be fabricated and cleaned for breathing air.

(3) Pressure rating: 6,000 psig

(4) Capacity: 500 cubic feet

(5) Cylinder valve, relief valve, and pressure gauge complying with ASME shall be installed.

(6) Two (2) cylinders on the front side shall be used for 6,000 psig system. The remaining two (2) cylinders on the back side shall be used for 3,000 psig system.

(7) Quantity: 4 ea

(8) Cylinders shall be commercial off-the-shelf items.

g. Cylinder stand

(1) Cylinder stand containing four (4) cylinders listed in Paragraph 4.f vertically shall be provided.

- (2) Cylinder stand shall be able to be fixed on the existing surface plate with T-slots.
- (3) Steel enclosure shall be provided. Clear polycarbonate window shall be provided for gauges. Openings shall be provided for valves and regulators to allow operation.
- (4) Commercial off-the-shelf items can be used with adding necessary modification.

h. Relief valve shall be installed in accordance with Enclosure (2).

- (1) Set pressure: 3,300 +150/-150 psig
- (2) Reseat pressure: 3,000 psig or higher
- (3) Purge flow rate: 400 L/min
- (4) Connection: NPT
- (5) Set pressure shall be adjustable.
- (6) Proposal model: R Series (Mfr: Swagelok) or equivalent

i. Regulator set to 4,000 psig shall be installed in accordance with Enclosure (2).

j. Pressure gauge shall be installed in accordance with Enclosure (2).

- (1) Range and unit: 0 to 10,000 psig
- (2) Dial size: 2.5 inches
- (3) Accuracy: $\pm 1.5\%$ of span or greater
- (4) Connection: NPT threads
- (5) Proposal model: B Model (Mfr: Swagelok) or equivalent

5. Procedure of replacement

a. The Government will depressurize the system.

b. The contractor shall disconnect the following piping connected to/from the existing quadruplet. Immediately install FME closures to the openings. Keep the FME closures attached until immediately before connecting.

- (1) 5,000 psig inlet piping from HPAC
- (2) 5,000 psig outlet piping to SCBA Maintenance Room
- (3) 5,000 psig outlet piping to Clean Room
- (4) 3,000 psig inlet piping from HPAC
- (5) 3,000 psig outlet piping to Clean Room

c. The Government will remove existing four (4) cylinders.

d. The contractor shall transport four (4) new cylinders and cylinder stand to a location near the installation site.

e. The Government will transport the cylinders and cylinder stand to the installation area using an overhead crane.

f. The contractor shall fix the cylinder stand on the existing surface plate. The contractor shall install four cylinders to the cylinder stand.

g. The contractor shall connect components in accordance with Enclosure (2). Each component shall be connected with $\Phi 1/4''$ ($\Phi 6.35$ mm) seamless stainless steel tube and Swagelok or equivalent tube fittings.

(1) Piping shall be appropriately looped and/or bent to relieve stress.

(2) Tubes shall be appropriately fixed.

6. Delivery and unloading

The contractor shall deliver and unload materials to the installation site in Bldg. A48.

7. Operational test

a. After installation, the contractor shall conduct operational tests to verify the system works properly.

b. The Government will operate HPACs and slowly pressurize the system to the working pressure.

c. The contractor shall check the proper operation of pressure gauges.

d. The contractor shall conduct a leak check.

e. The contractor shall purge residual air in the system and fill the system with air supplied by HPACs.

f. Technical Point of Contact (TPOC) will witness the operational test.

8. Document submission

The contractor shall submit the following documents. Unless otherwise specified, documents shall be written in Japanese or English.

a. Pre-fabrication documents

The contractor shall submit the following documents to TPOC for review and approval before the start of work, no later than twenty-five (25) working days after the contract award. The Government technical review will be completed and returned with comments within ten (10) working days after receiving from the contractor.

(1) Sketch of the system

(2) Connection diagram

(3) Parts list with cleanliness information

(4) Work schedule

(5) Catalog materials for commercial off-the-shelf items

(6) Lifting plan for cylinders and cylinder stand

b. Completion Documents

The contractor shall submit the following completion documents by three (3) Japanese paper copies, one (1) English paper copy, and one (1) CD/DVD media containing Japanese and English files before the delivery.

(1) Report on the entire work

(2) Final drawing

(3) Parts list with cleanliness information

(4) Technical manuals for commercial off-the-shelf items

(5) Pre-shipping test certificates for accuracy of pressure gauges

(6) Pre-shipping test certificates for set and reseal pressure of relief valves

- (7) Result of operational test
- (8) Warranty sheet for one year

9. TPOC

To be determined (TBD)